

Final Report:

NFL Matchup Prediction Analysis

1. Problem Statement

This project aims to identify natural groupings of fish species based on when and where they are caught in commercial groundfish operations. Using unsupervised machine learning, species are clustered by co-occurrence patterns across statistical areas and seasons to reveal ecological or operational relationships. These insights can support ecosystem-based management, predict bycatch risks, and inform seasonal or spatial strategies for fisheries management.

2. The Data

Observer data was obtained from NOAA's Northeast Fisheries Science Center through the Northeast ASM program. It comprises over 4 million records of species and debris per haul from commercial groundfish vessels between 1999 and 2025. Each record includes the year and month of landing, location by quarter-degree latitude/longitude square (QDSQ), fishing gear type, and whether the haul was observed. The dataset also provides species-level details, including common and scientific names, species ID, live weights, and flags indicating whether the catch was weighed, dressed or round.

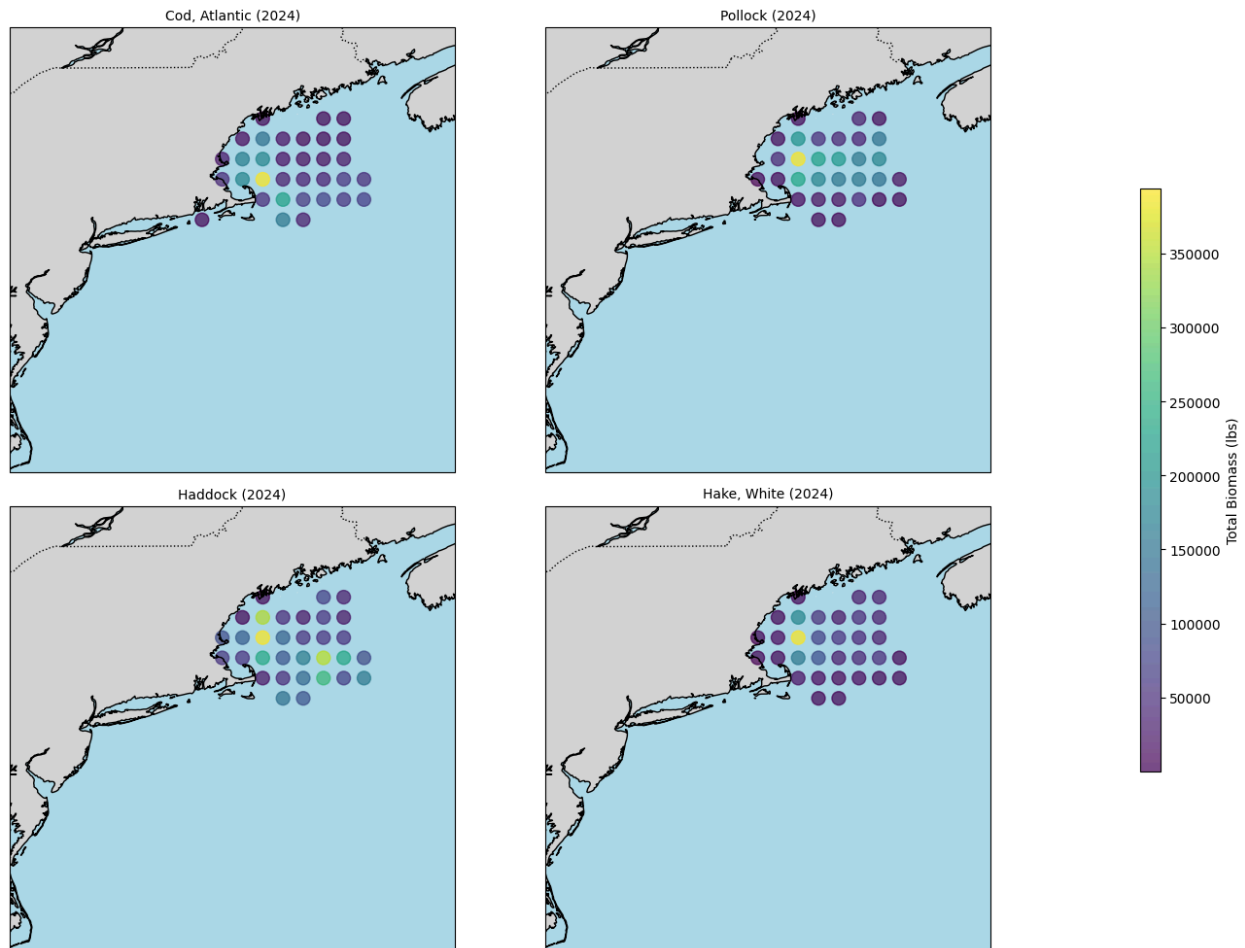
While not all hauls were observed and coverage varies across years, the dataset's combination of temporal, spatial, and species-level information provides a detailed foundation for analyzing species co-occurrence and community patterns. Some rare species may be filtered during analysis, and spatial resolution is limited to QDSQs, with potential sampling biases based on gear type.

3. Data Wrangling

For this analysis, we focused on ASM-managed species as well as other commonly encountered species to create a subset that captured the majority of ecological and operational interactions. This filtering step helped reduce noise from rarely observed species and made the clustering results more interpretable. Additional preprocessing included standardizing columns, imputing missing values, removing unobserved hauls, and converting QDSQ to approximate lat/lon.

4. Exploratory Data Analysis

We explored the dataset by visualizing biomass by species to understand general patterns and relative contributions. This allowed us to identify which species consistently accounted for the largest portion of total biomass and to focus on the subset of species most relevant for the analysis based on domain knowledge. These observations provided context for the later clustering, helping to ensure that the analysis emphasized species with meaningful ecological and operational presence.



5. Modeling and Findings

We explored both hierarchical and k-means clustering to identify species groupings, evaluating each approach using the silhouette score, which measures how well points fit within their assigned clusters relative to others. To prepare the data for clustering, haul-level records were aggregated into spatiotemporal strata unique combinations of year, month, and location (QDSQ) and biomass was summarized by species within each stratum. This transformation produced a structured, model-ready input that captured co-occurrence patterns across time and space.

While both approaches provided meaningful groupings, k-means outperformed hierarchical clustering in terms of silhouette score. Testing multiple values of k , $k = 5$ yielded a silhouette score of 0.246, while $k = 8$ improved the score to 0.277 and was selected as the final model. Validation on a holdout dataset produced a silhouette score of 0.254, indicating that the cluster structure generalizes well to unseen data. Resulting clusters were reality-checked with domain knowledge and visualized using Cartopy to show seasonal patterns of cluster occurrence, confirming that the groupings were consistent with known ecological and operational relationships.

6. Next Steps

The clusters identified in this analysis can be used to evaluate current fishing practices, including seasonal closures, gear restrictions, and spatial regulations, to ensure they align with observed species co-occurrence patterns. They can also inform quota allocations and bycatch limits, helping to ensure management measures reflect ecological realities. Additionally, clusters can highlight regions or seasons where sensitive species co-occur with target species, identifying potential bycatch hotspots. Overall, these insights support ecosystem-based management by guiding multispecies strategies and planning for sustainable fisheries.

